

Lab 1: Diodes and Rectifiers

ECE 0102: Microelectronic Circuits + Lab

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Stefan Zhang.....Student ID# 2024141520349

Knight LiuStudent ID# 2024141520361

Team Roles:

Activity	Student Name
Circuit Construction	Stefan Zhang
Data Collection	Knight Liu
Data Analysis	Stefan Zhang; Knight Liu
Answers to Questions	Stefan Zhang; Knight Liu
Lab Report Writing	Stefan Zhang; Knight Liu

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1) Objectives and Background

The objective of this lab was to study the electrical characteristics of a silicon diode and to understand its application in a rectifier circuit. In Action 1.1, the current-voltage (I-V) characteristics of the diode were measured under different DC input voltages, and the diode parameters were estimated from the measured data. In Action 2, the rectifying behavior of the diode was observed by applying a sinusoidal input signal and comparing the input and output waveforms. This lab demonstrates both the nonlinear behavior of a pn-junction diode and its practical use in basic rectifier circuits.

2) Equipment Used

Table 1: Equipments Used in Lab 1

Item	Quantity
1N4007 Silicon Rectifier Diode	1
Digital Multimeter	2
DC Power Supply	1
Function Generator	1
Oscilloscope	1
Resistor ($100\ \Omega$, measured as $99.831\ \Omega$)	1
Resistor ($10\ \text{k}\Omega$)	1

3) Action 1.1: Diode I-V Characteristics

3.1) Procedure

The circuit for Action 1.1 was built using one 1N4007 diode and one resistor. The measured resistor value was

$$R = 99.831 \, \Omega$$

A DC power supply was used to set V_{IN} from -5 V to 5 V . For each input voltage, the diode voltage V_D was measured, and the diode current was calculated by

$$I_{D1} = \frac{V_{IN} - V_D}{R}$$

3.2) Results and Analysis

The measured results show the typical nonlinear behavior of a silicon diode. Under reverse bias, the current was nearly zero. Under forward bias, the current remained very small at low voltage and then increased rapidly when the diode voltage approached about 0.6 V .

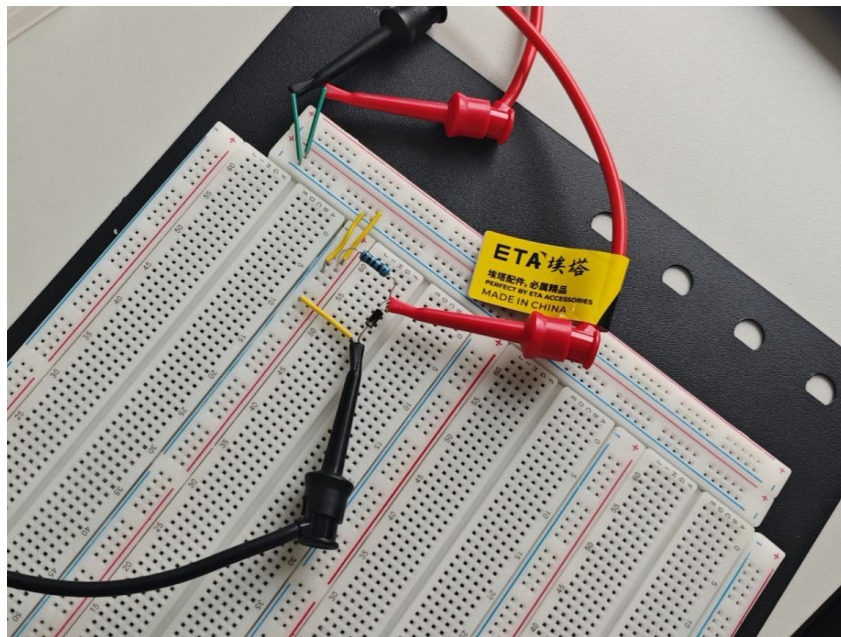


Figure 1: Experimental setup for Action 1.1.

Table 2: Measured data for Action 1.1 and calculated results for Action 1.2

V_{IN} [V]	V_D [V]	I_{D1} [A]	I_{D2} [A]	Error [%]
-5.0	-4.999	≈ 0	—	—
0	-0.000931	9.33×10^{-6}	—	—
0.1	0.099	1.00×10^{-5}	1.34×10^{-6}	—
0.2	0.198	2.00×10^{-5}	1.16×10^{-6}	—
0.3	0.298	2.00×10^{-5}	1.03×10^{-6}	—
0.4	0.398	2.00×10^{-5}	9.38×10^{-6}	53.20
0.5	0.492	8.01×10^{-5}	8.04×10^{-5}	0.30
0.6	0.561	3.91×10^{-4}	3.89×10^{-4}	0.39
0.7	0.603	9.72×10^{-4}	1.02×10^{-3}	4.59
0.8	0.629	1.71×10^{-3}	1.84×10^{-3}	7.49
0.9	0.646	2.54×10^{-3}	2.72×10^{-3}	6.73
1.0	0.657	3.44×10^{-3}	3.49×10^{-3}	1.63
2.0	0.715	1.29×10^{-2}	1.31×10^{-2}	2.13
5.0	0.768	4.24×10^{-2}	4.41×10^{-2}	4.15
Average error for V_{IN} from 0.4 to 5 V				8.96

4) Action 1.2: Extraction of Diode Parameters

4.1) Method

The diode current was modeled by

$$I_D = I_S \exp\left(\frac{V_D}{nV_T}\right)$$

with

$$V_T = 25 \text{ mV}$$

Taking logarithm gives

$$\ln I_D = \ln I_S + \frac{V_D}{nV_T}$$

Thus, I_S and n can be extracted from the forward-bias data.

4.2) Results and Analysis

From the fitting result, the extracted parameters were

$$I_S \approx 1.54 \times 10^{-9} \text{ A}$$

$$n \approx 1.83$$

The model agrees reasonably well with the measured data in the forward conduction region. The error is larger near the turn-on region, where the current is very small and measurement uncertainty is more significant.

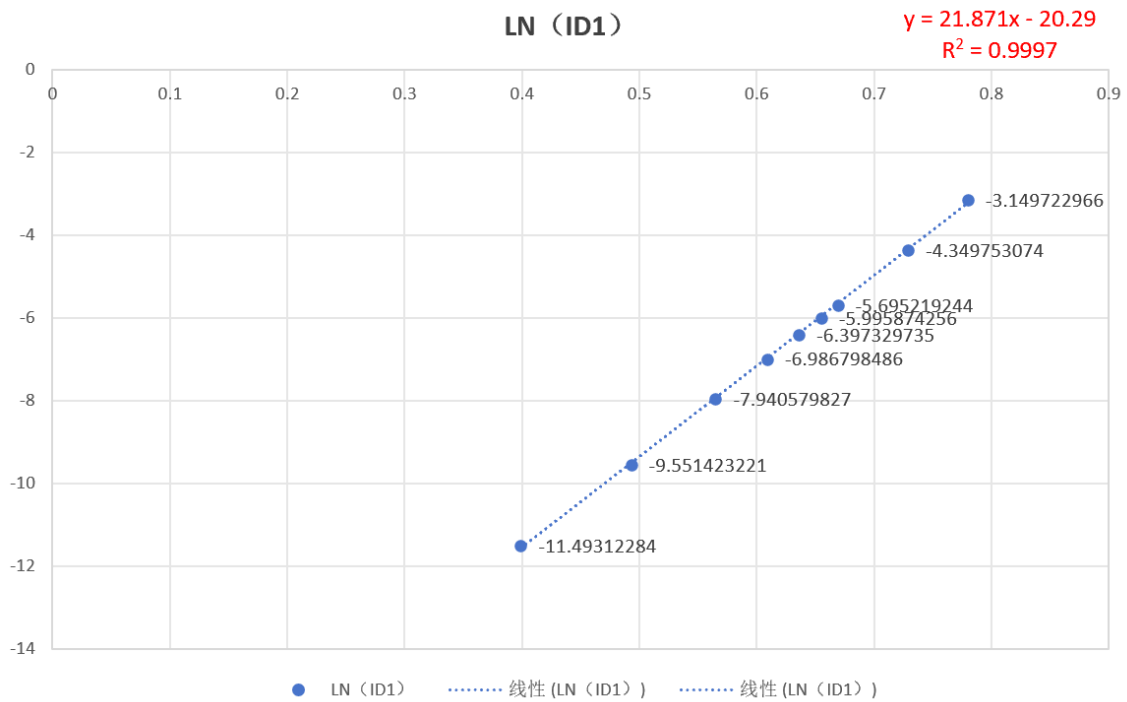


Figure 2: Fitting result for Action 1.2.

Table 3: Extracted diode parameters

Parameter	Value
R	99.831Ω
I_S	$1.54 \times 10^{-9} \text{ A}$
n	1.83

5) Action 2.1: Time-Domain Measurement of the Rectifier Circuit

5.1) Procedure

The rectifier circuit was built using one diode and one 10 k Ω resistor.

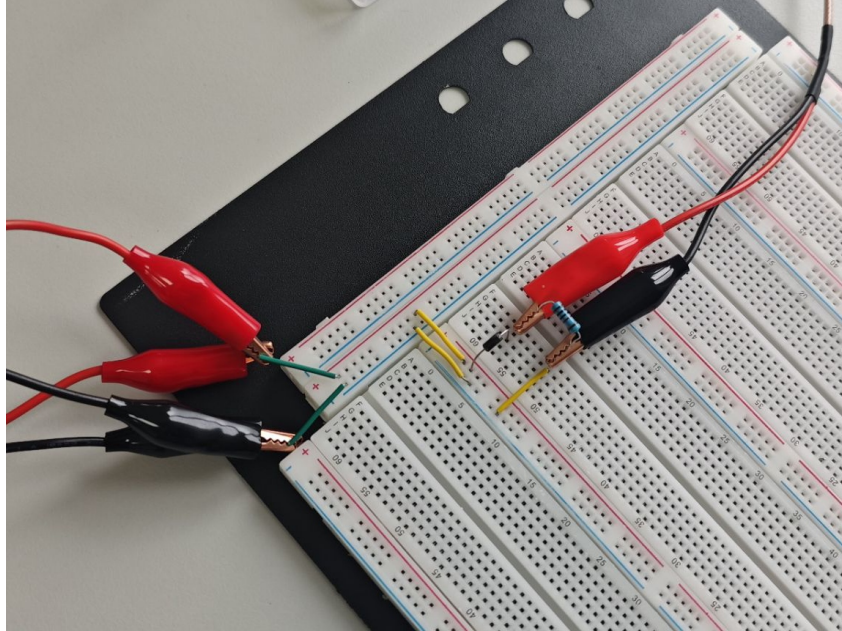


Figure 3: Experimental setup for Action 2.1.

A sinusoidal input signal was applied with

$$V_{IN} = 10 \text{ V}_{PP}, \quad f = 1 \text{ kHz}$$

The oscilloscope was used to observe both $V_{IN}(t)$ and $V_{OUT}(t)$.

5.2) Results and Analysis

The output waveform showed half-wave rectification. During the negative half cycle, the diode was reverse biased and the output remained close to zero. During the positive half cycle, the output followed the input after the diode turned on.



Figure 4: Measured input and output waveforms in Action 2.1.

6) Action 2.2: Transfer Characteristics of the Rectifier Circuit

6.1) Procedure

The oscilloscope was switched to X-Y mode, with V_{IN} as the horizontal axis and V_{OUT} as the vertical axis.

6.2) Results and Analysis

The measured transfer curve remained near zero when the input voltage was below the diode turn-on voltage. When the input exceeded the threshold, the output increased approximately linearly. The transfer characteristics can be written as

$$V_{OUT} = \begin{cases} 0, & V_{IN} \leq V_{\gamma} \\ V_{IN} - V_{\gamma}, & V_{IN} > V_{\gamma} \end{cases}$$

where $V_{\gamma} \approx 0.7 \text{ V}$.

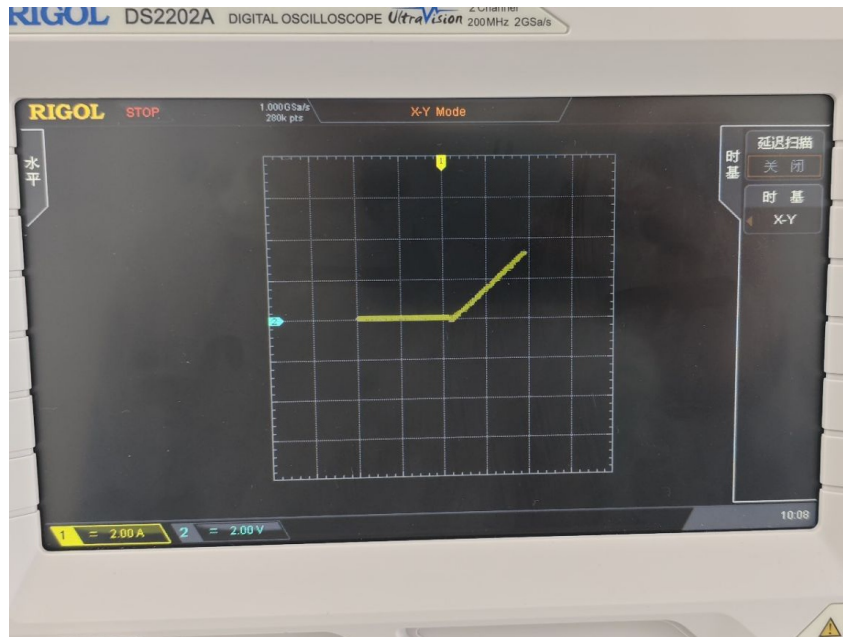


Figure 5: Transfer characteristics measured in Action 2.2.

7) Answers to Post-Lab Questions

Question 1

The diode shows nonlinear behavior because the current does not change linearly with voltage. It remains very small at low forward voltage and then increases rapidly after turn-on.

Question 2

At high forward current, only a small increase in voltage is needed to produce a large increase in current because the I-V relation is exponential.

Question 3

The difference between measured and fitted data is caused by non-ideal diode behavior, resistor tolerance, temperature effect, and measurement error.

Question 4

The rectifier output appears only during one half cycle because the diode conducts only in forward bias and blocks current in reverse bias.